

(12) **United States Patent**
Ali et al.

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(54) **METHOD AND SYSTEM FOR ANTENNA
ARRAY CALIBRATION FOR
CROSS-COUPLING AND GAIN/PHASE
VARIATIONS IN RADAR SYSTEMS**

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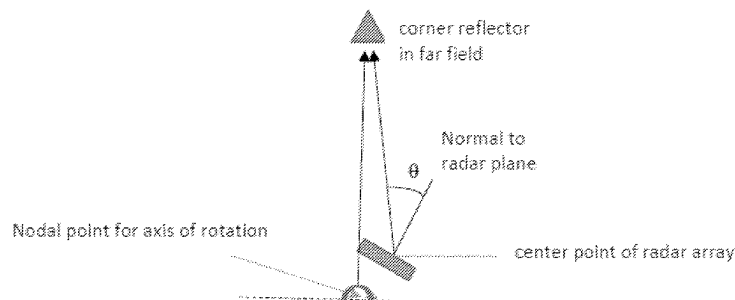
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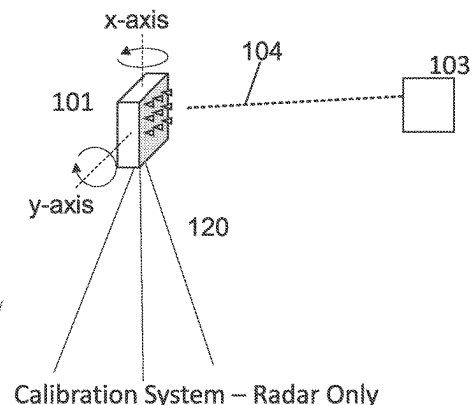
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(57) **ABSTRACT**

A radar system with on-system calibration includes capa-
bilities for radar detection and correction for system impair-
ments to improve detection performance. The radar system
is equipped with pluralities of transmit antennas and plu-
ralities of receive antennas. The radar system uses a series of
calibration measurements of a known object to estimate the
system impairments. A correction is then applied to the
beamforming weights to mitigate the effect of these impair-
ments on radar detection. The estimation and correction
requires no external measurement equipment and can be
computed on the radar system itself.

26 Claims, 10 Drawing Sheets



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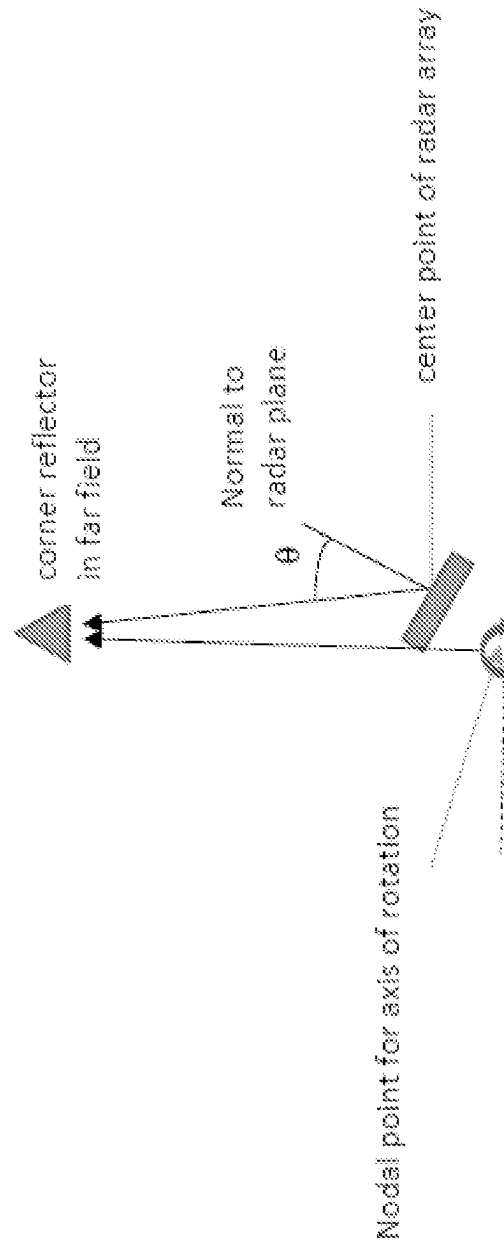


FIG. 1B

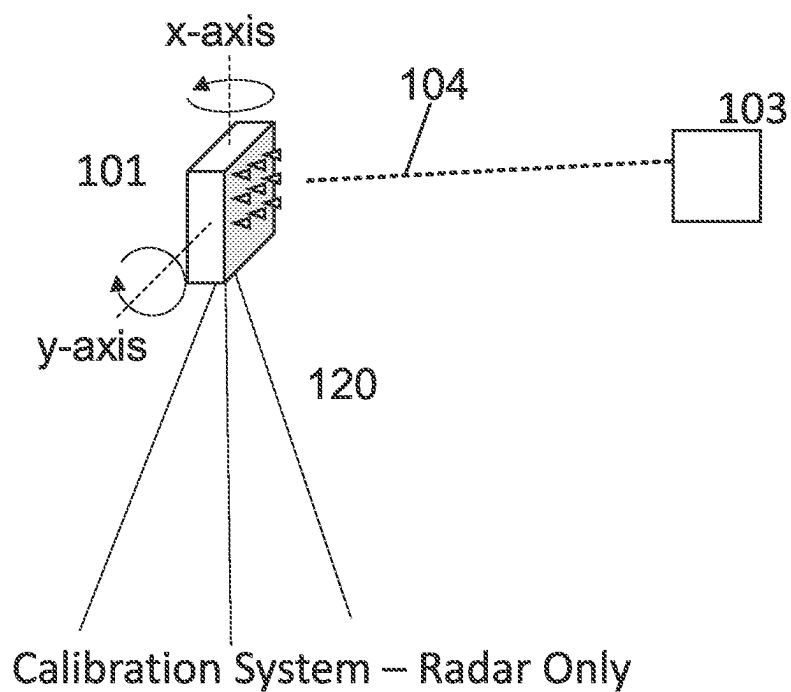
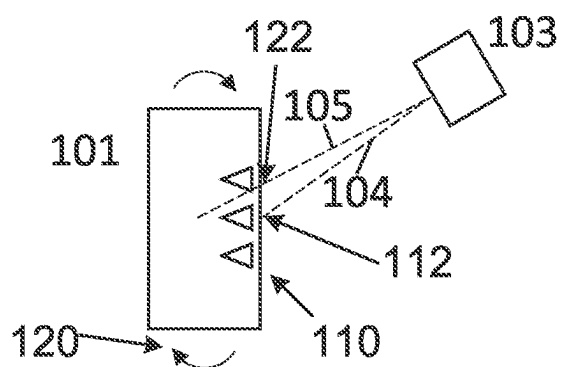
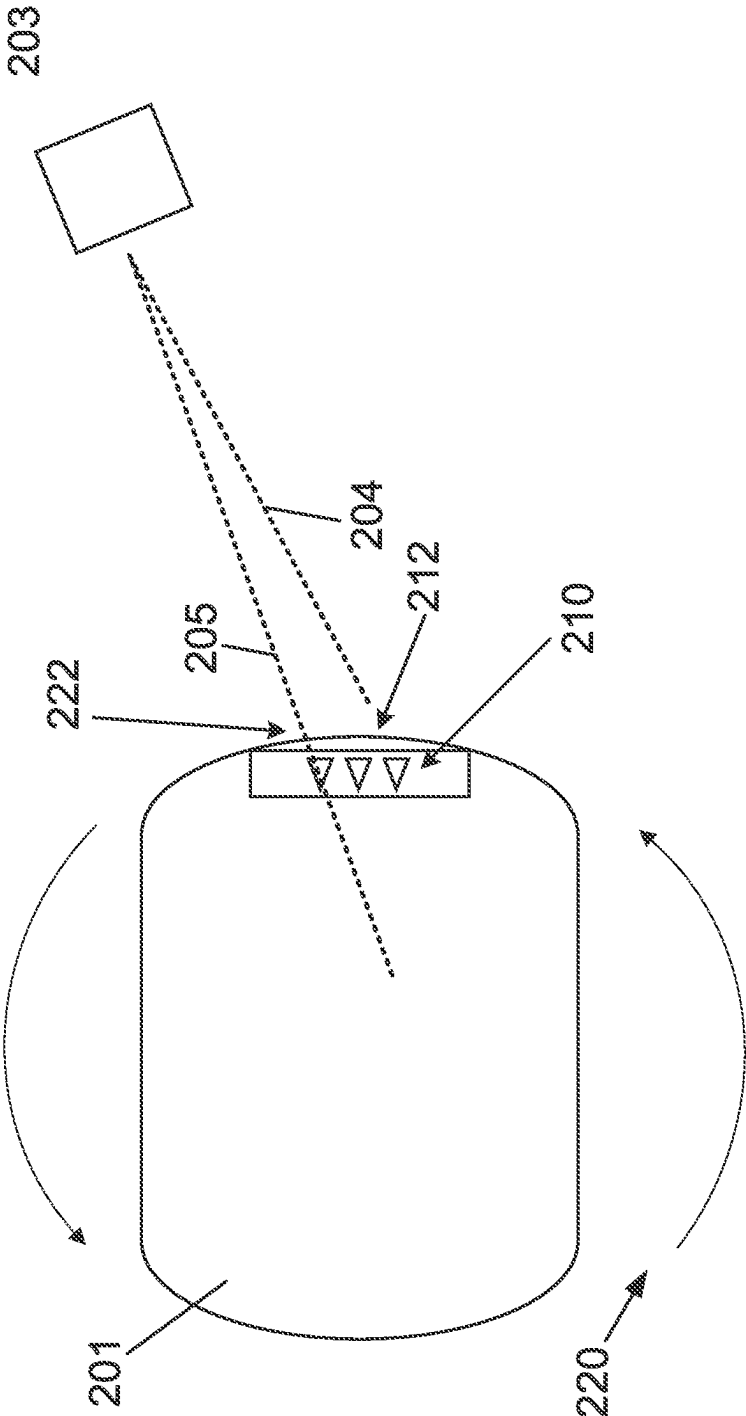


FIG. 1C



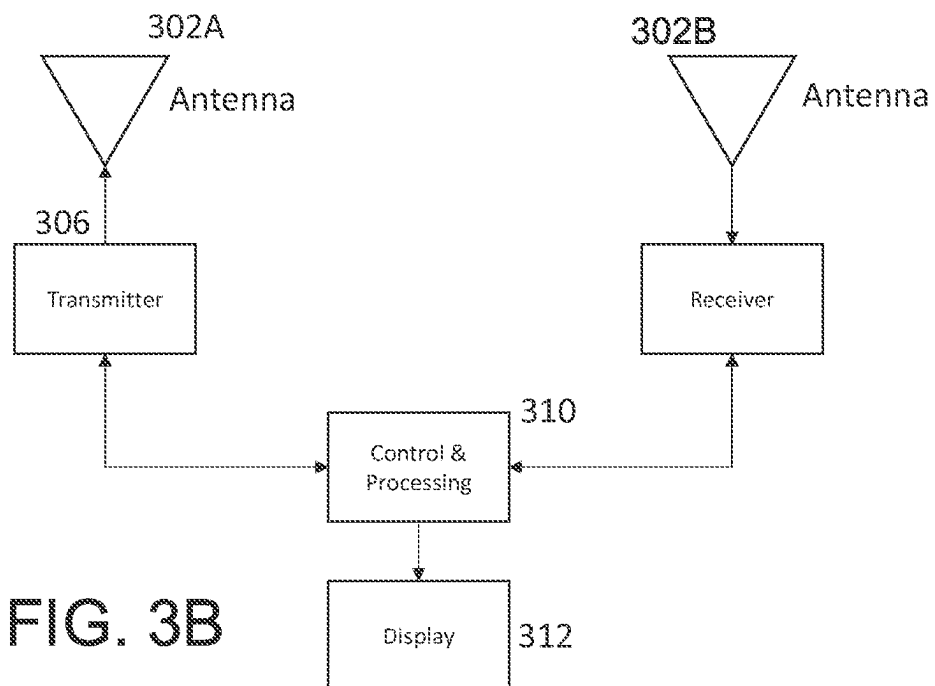
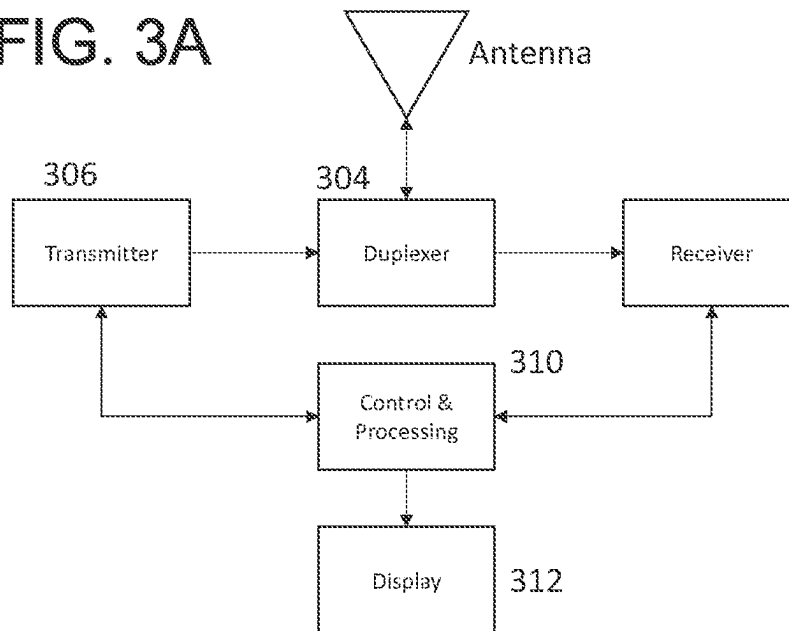
When nodal point for axis of rotation
does not match the radar array center

FIG. 2



Calibration System – Radar On Platform

FIG. 3A



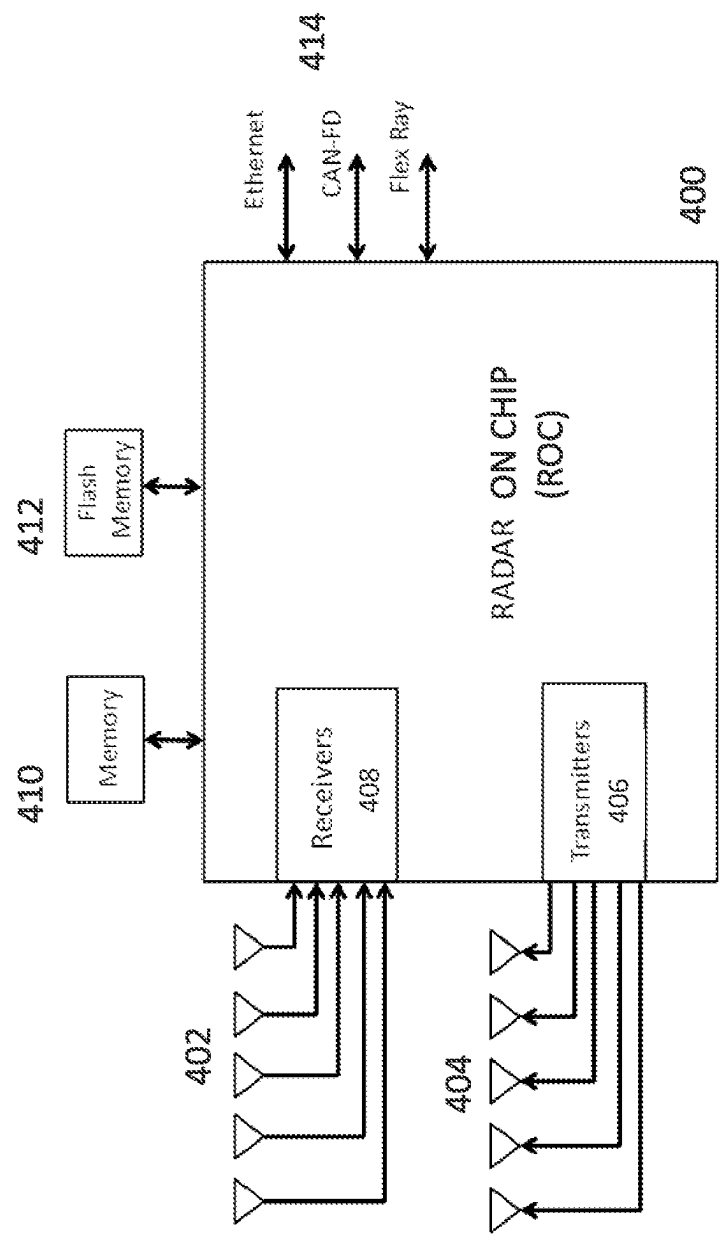
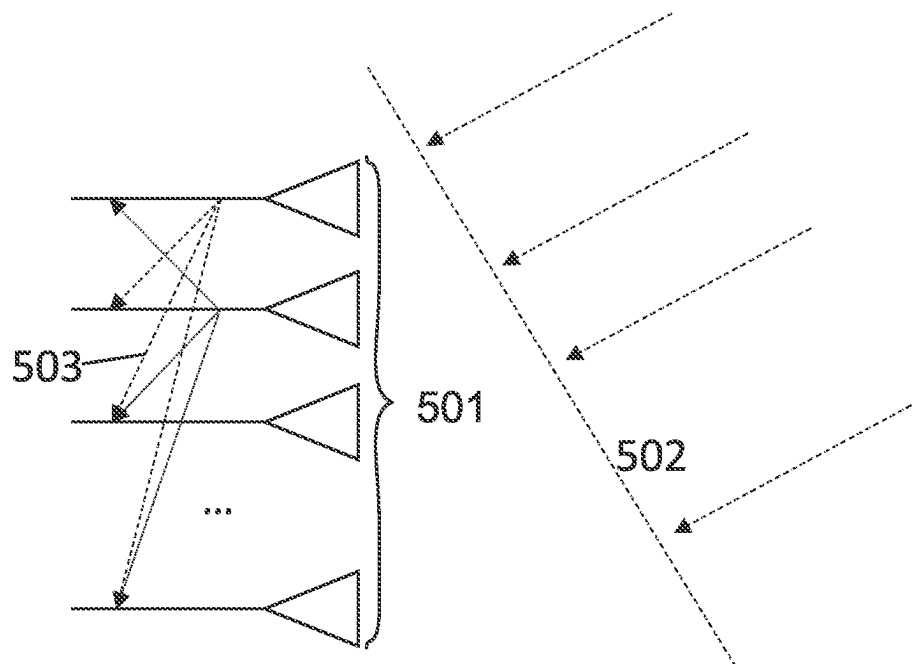


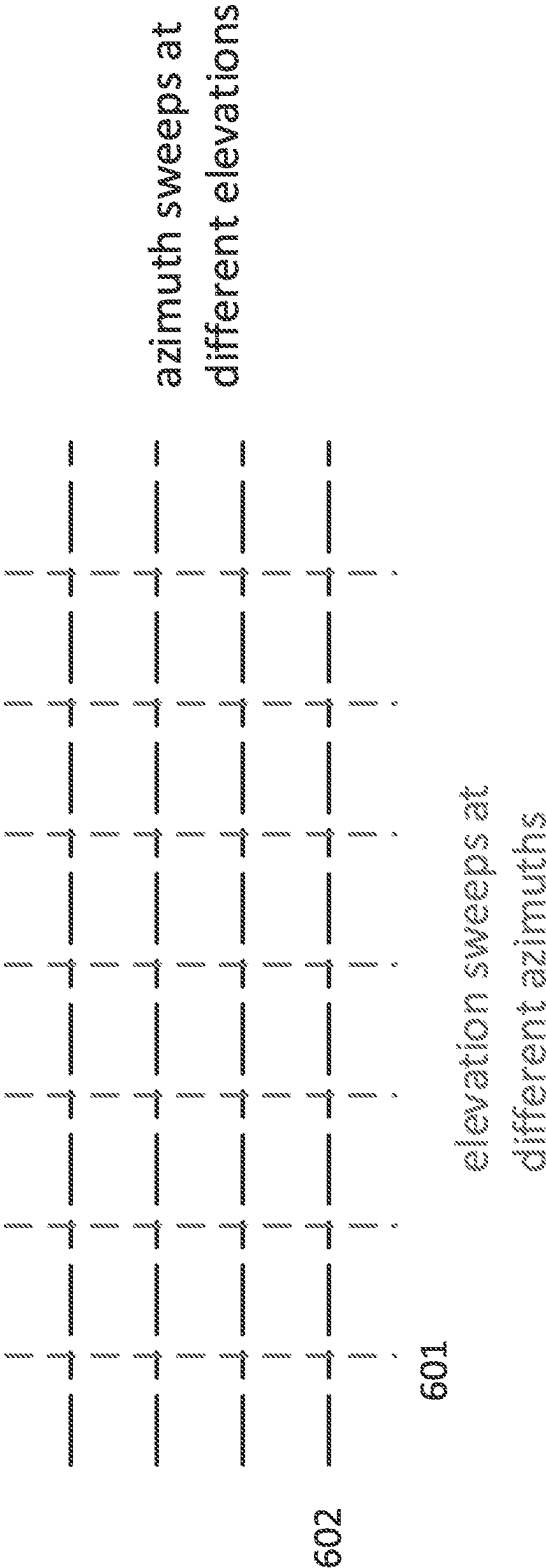
FIG. 4

FIG. 5



Cross Coupling

FIG. 6



Sweep Patterns

FIG. 7

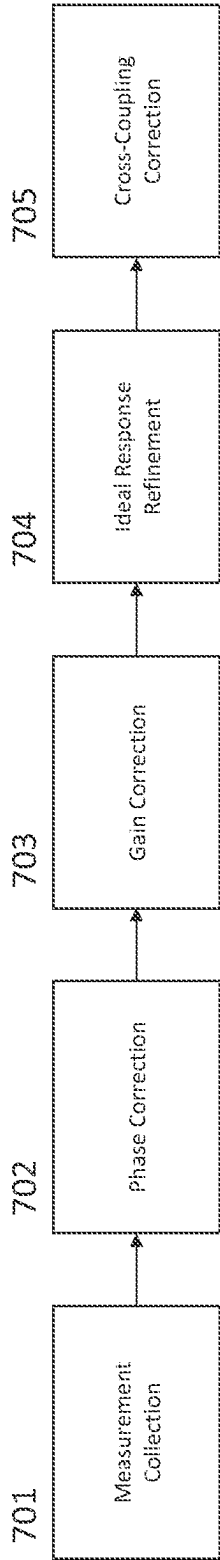


FIG. 8: Phase Correction

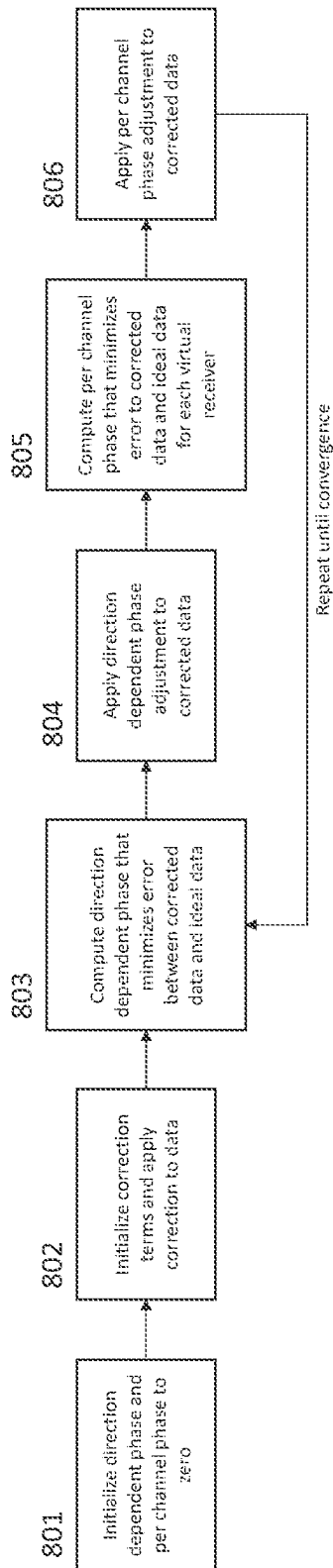


FIG. 9: Gain Correction

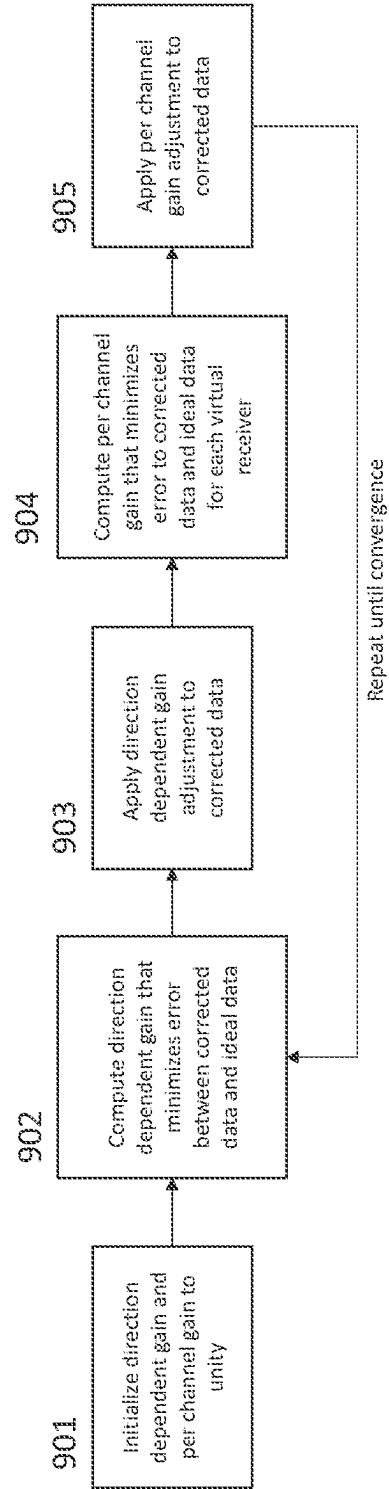
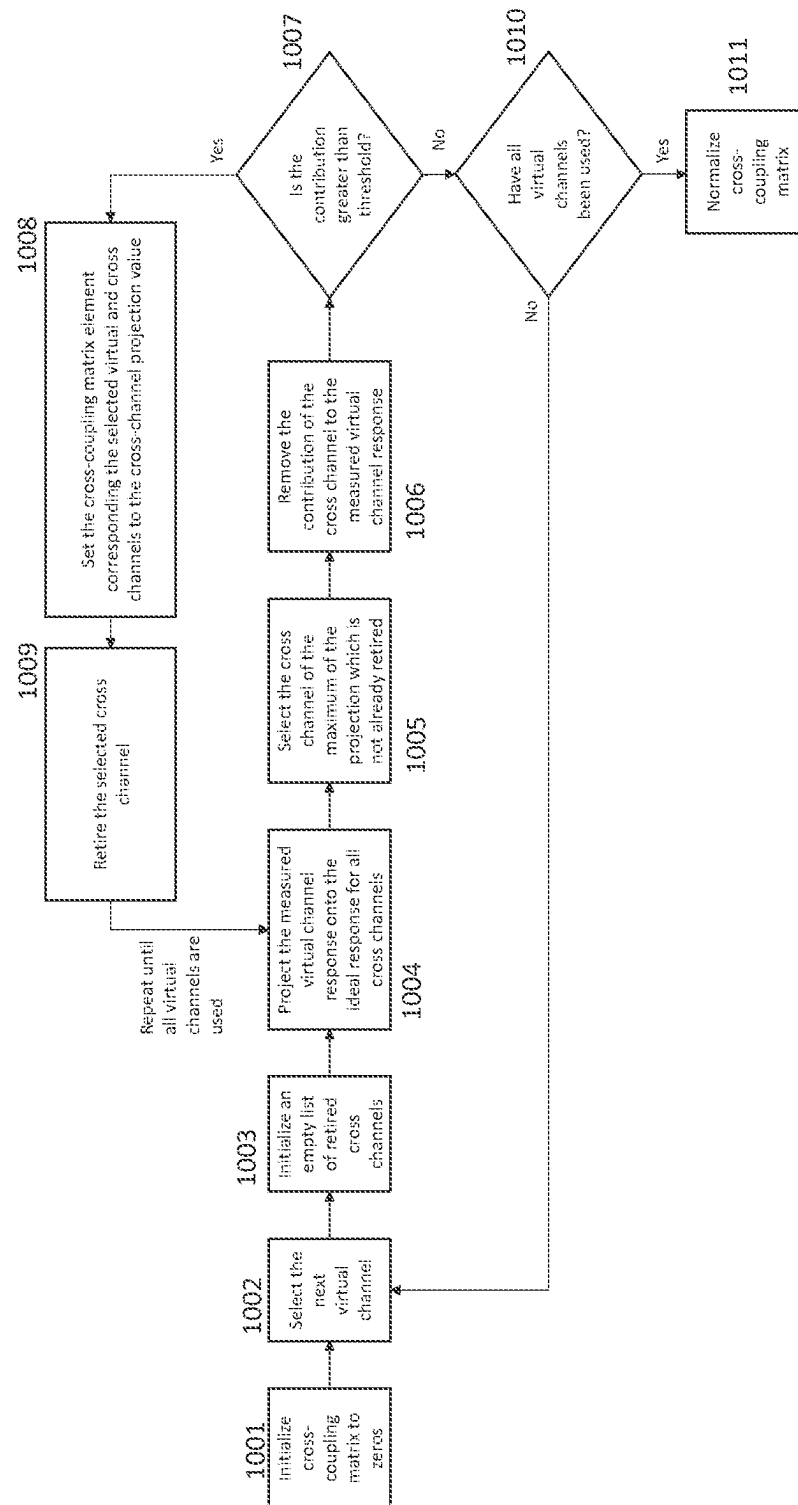


FIG. 10: Cross-Coupling Correction



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METHOD AND SYSTEM FOR ANTENNA ARRAY CALIBRATION FOR CROSS-COUPLING AND GAIN/PHASE VARIATIONS IN RADAR SYSTEMS

CROSS REFERENCE TO RELATED APPLICATION

The present application claims priority to and is a continuation of U.S. patent application Ser. No. 17/147,914, filed Jan. 13, 2021, which claims the benefits of U.S. provisional application, Ser. No. 62/960,220, filed Jan. 13, 2020, which are hereby incorporated by reference herein in their entireties.

The present invention is directed to radar systems, and more particularly to radar systems for vehicles and robotics.

BACKGROUND OF THE INVENTION

The use of radar to determine location, range, and velocity of objects in an environment is important in a number of applications including automotive radar, industrial processes, robotic sensing, gesture detection, and positioning. A radar system typically transmits radio signals and listens for the reflection of the radio signals from objects in the environment. By comparing the transmitted radio signals with the received radio signals, a radar system can determine the distance to an object, and the velocity of the object. Using multiple transmitters and/or receivers, or a movable transmitter or receiver, the location (angle) of an object can also be determined. Therefore, radar systems require accurate operation to maintain their optimal performance.

SUMMARY OF THE INVENTION

Embodiments of the present invention provide for a radar calibration system that calibrates for radar system impairments using a series of radar data measurements. Such impairments include coupling effects, per channel gain and phase variations, and direction dependent gain and phase variations. This calibration system operates under a variety of environments, with a variety of external information, and with a variety of objective functions to modify the measurement collection as well as the calibration processing to optimize the system with respect to a given objective function.

In an aspect of the present invention, a radar system for a robot or vehicle that calibrates for system impairments includes a radar system with at least one transmitter and at least one receiver. The transmitter and receiver are connected to at least one antenna. The transmitter is configured to transmit radio signals. The receiver is configured to receive a radio signal that includes the transmitted radio signal transmitted by the transmitter and reflected from objects in the environment. The receiver is also configured to receive radio signals transmitted by other radar systems.

In an aspect of the present invention, the radar system comprises one of: a single transmitter and a plurality of receivers; a plurality of transmitters and a single receiver; and a plurality of transmitters and a plurality of receivers.

In a further aspect of the present invention, the transmitters and receivers may be connected to multiple antennas through a switch.

In another aspect of the present invention, the radar system includes a calibration module that is configured to rotate its direction in both azimuth and elevation. In the presence of at least one reflecting object, the calibration

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module collects reflected signals from the at least one reflecting object at desired angles of interest in the azimuth and elevation space. This rotation may occur in either a continuous manner or a discrete “stop-and-go” manner. The radar system’s center point of the antenna array does not need to align with the center point of rotation, and the radar system corrects for phase distortion and angle-of-arrival error due to this misalignment. This misalignment is referred to as nodal displacement. The calibration module then processes these measurements into a correction matrix, which calibrates for radar system impairments. These may include phase error due to nodal displacement, per channel phase variation, direction dependent phase variation, per channel amplitude variation, direction dependent amplitude variation, and channel response cross coupling. The angles-of-arrival of the collected reflected signals may be either estimated by the radar system or determined through prior knowledge of the object(s) location(s) relative to the radar system.

In another aspect of the present invention, the radar system may modify its measurement collection and calibration processing to optimize different objective functions. These modifications include the speed and manner of rotation, quantity of measurements collected, and the selection of antenna(s) and channel(s) transmitting and receiving the signal(s). These modifications also include parameters in the processing that control the computation of the correction matrix and affect the processing speed and correction accuracy.

In another aspect of the present invention, a method for calibrating a radar system for system impairments includes at least one transmitter transmitting radio signals. At least one receiver is receiving radio signals that include radio signals transmitted by the transmitter and reflected from objects in an environment. The at least one transmitter and the at least one receiver are coupled to an antenna array. A platform rotating the at least one receiver and the at least one transmitter in both azimuth and elevation. An array center of the antenna array is not aligned with the platform’s rotational center. The method includes collecting, with a calibration module, in the presence of at least one object, reflected signals from the at least one object at desired angles of interest in azimuth and elevation, calculating a misalignment between the array center of the antenna array and the rotation center of the platform. The method also includes correcting, with the at least one receiver, for phase distortion and angle-of-arrival error due to the calculated misalignment. The misalignment between the array center of the antenna and the rotation center of the platform is a nodal displacement. The array center of the antenna array is a nodal point.

These and other objects, advantages, purposes and features of the present invention will become apparent upon review of the following specification in conjunction with the drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1A is a diagram of a radar system where a nodal point for an axis of rotation does not match a radar array center of the radar system in accordance with the present invention;

FIG. 1B is a perspective view of a radar calibration system orientated towards a target in accordance with the present invention;

FIG. 1C is another view of the radar calibration system and target of FIG. 1B;

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FIG. 2 is a diagram of a radar calibration system installed on a maneuverable platform in accordance with the present invention;

FIG. 3A and FIG. 3B are block diagrams of radar systems that use the calibration system in accordance with the present invention;

FIG. 4 is a block diagram illustrating a radar with a plurality of receivers and a plurality of transmitters (MIMO radar) that uses the calibration system in accordance with the present invention;

FIG. 5 is a visualization of an exemplary antenna array, an exemplary received plane wave, and exemplary coupling effects in accordance with the present invention;

FIG. 6 is a diagram of exemplary sweep patterns executed during an exemplary measurement procedure in accordance with the present invention;

FIG. 7 is a flow chart describing the high-level processes of the calibration procedure, in accordance with the present invention;

FIG. 8 is a flow chart describing the process of estimating direction dependent and per channel phase correction, in accordance with the present invention;

FIG. 9 is a flow chart describing the process of estimating direction dependent and per channel gain correction, in accordance with the present invention; and

FIG. 10 is a flow chart describing the process of estimating cross-coupling correction, in accordance with the present invention;

DESCRIPTION OF THE PREFERRED EMBODIMENTS

Referring to the drawings and the illustrative embodiments depicted therein, wherein numbered elements in the following written description correspond to like-numbered elements in the figures, a calibration system provides for a calibration of a radar system. The radar system includes a calibration module that includes a platform for rotating receivers and transmitters of the radar system in both azimuth and elevation. An array center of the antenna array is not aligned with the platform's rotational center. The calibration module collects, in the presence of at least one object, reflected signals from the at least one object at desired angles of interest in azimuth and elevation. The calibration module calculates a misalignment between the array center of the antenna array and the rotation center of the platform. The at least one receiver corrects for phase distortion and angle-of-arrival error due to the calculated misalignment. The misalignment between the array center of the antenna and the rotation center of the platform is a nodal displacement. The array center of the antenna array is a nodal point.

An exemplary radar system operates by transmitting one or more signals from one or more transmitters and then listening for reflections of those signals from objects in the environment by one or more receivers. By comparing the transmitted signals and the received signals, estimates of the range, velocity, and angle (azimuth and/or elevation) of the objects can be estimated.

There are several different types of signals that transmitters in radar systems employ. A radar system may transmit a pulsed signal or a continuous signal. In a pulsed radar system, the signal is transmitted for a short time and then no signal is transmitted. This is repeated over and over. When the signal is not being transmitted, the receiver listens for echoes or reflections from objects in the environment. Often a single antenna is used for both the transmitter and receiver

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and the radar transmits on the antenna and then listens to the received signal on the same antenna. This process is then repeated. In a continuous wave radar system, the signal is continuously transmitted. There may be an antenna for transmitting and a separate antenna for receiving.

Another classification of radar systems is the modulation of signal being transmitted. A first type of continuous wave radar signal is known as a frequency modulated continuous wave (FMCW) radar signal. In an FMCW radar system, the transmitted signal is a sinusoidal signal with a varying frequency. By measuring a time difference between when a certain frequency was transmitted and when the received signal contained that frequency, the range to an object can be determined. By measuring several different time differences between a transmitted signal and a received signal, velocity information can be obtained.

A second type of continuous wave signal used in radar systems is known as a phase modulated continuous wave (PMCW) radar signal. In a PMCW radar system, the transmitted signal from a single transmitter is a sinusoidal signal in which the phase of the sinusoidal signal varies. Typically, the phase during a given time period (called a chip period or chip duration) is one of a finite number of possible phases. A spreading code consisting of a sequence of chips, (e.g., +1, +1, -1, +1, -1 . . .) is mapped (e.g., +1→0, -1→□) into a sequence of phases (e.g., 0, 0, □, 0, □ . . .) that is used to modulate a carrier to generate the radio frequency (RF) signal. The spreading code could be a periodic sequence or could be a pseudo-random sequence with a very large period so it appears to be a nearly random sequence. The spreading code could be a binary code (e.g., +1 or -1). The resulting signal has a bandwidth that is proportional to the rate at which the phases change, called the chip rate R_c , which is the inverse of the chip duration, $T_c=1/R_c$. By comparing the return signal to the transmitted signal, the receiver can determine the range and the velocity of reflected objects.

In some radar systems, the signal (e.g. a PMCW signal) is transmitted over a short time period (e.g. 1 microsecond) and then turned off for a similar time period. The receiver is only turned on during the time period where the transmitter is turned off. In this approach, reflections of the transmitted signal from very close targets will not be completely available because the receiver is not active during a large fraction of the time when the reflected signals are being received. This is called pulse mode.

Digital frequency modulated continuous wave (FMCW) and phase modulated continuous wave (PMCW) are techniques in which a carrier signal is frequency or phase modulated, respectively, with digital codes using, for example, GMSK. Digital FMCW radar lends itself to be constructed in a MIMO variant in which multiple transmitters transmitting multiple codes are received by multiple receivers that decode all codes.

The advantage of the MIMO digital FMCW radar is that the angular resolution is that of a virtual antenna array having an equivalent number of elements equal to the product of the number of transmitters and the number of receivers. Digital FMCW MIMO radar techniques are described in U.S. Pat. Nos. 9,989,627; 9,945,935; 9,846,228; and 9,791,551, which are all hereby incorporated by reference herein in their entireties.

The radar sensing system of the present invention may utilize aspects of the radar systems described in U.S. Pat. Nos. 10,261,179; 9,971,020; 9,954,955; 9,945,935; 9,869,762; 9,846,228; 9,806,914; 9,791,564; 9,791,551; 9,772,397; 9,753,121; 9,689,967; 9,599,702; 9,575,160, and/or 9,689,967, and/or U.S. Publication Nos. US-2017-0309997;

and/or U.S. patent application Ser. No. 16/674,543, filed Nov. 5, 2019, Ser. No. 16/259,474, filed Jan. 28, 2019, Ser. No. 16/220,121, filed Dec. 14, 2018, Ser. No. 15/496,038, filed Apr. 25, 2017, Ser. No. 15/689,273, filed Aug. 29, 2017, Ser. No. 15/893,021, filed Feb. 9, 2018, and/or Ser. No. 15/892,865, filed Feb. 9, 2018, and/or U.S. provisional application, Ser. No. 62/816,941, filed Mar. 12, 2019, which are all hereby incorporated by reference herein in their entireties.

Antenna Calibration:

Determining a correct angle calibration matrix to counter the impact of effective cross-coupling between virtual receivers in large-scale MIMO systems has been challenging. The problem is especially acute when the system is large or cannot be conveniently placed on the rotating measurement system. In some cases, a nodal point cannot be maintained or cannot even be accurately determined. Such cases occur in radars mounted on robots, drones or other devices, or in cases when angle calibration is desired in situ with the whole system assembled. An exemplary method is disclosed that efficiently and correctly determines channel-to-channel variations and cross-coupling coefficients from angle sweep data in the presence of an unknown nodal point of the system. An exemplary algorithm also produces the diagonal calibration values as a by-product.

Typical angle calibration methods require collection of channel response data for a number of angles, which is also called as angle sweep data. The data is collected in an anechoic chamber with a single target in far-field and radar mounted on a gimbal that can be rotated between the angles of interest (up-to ± 90 degrees), which allows collecting the target virtual channel response in those angles. A typical data collection system is shown in FIG. 1A. This represents the case where the nodal point for the axis of rotation is the same as the center of the radar antenna system. The radar may have planar antenna array (2-D) instead of a linear antenna array (1-D). The radar will then need to rotate in two axes maintaining the nodal point of rotation in both axes at the center of the planar antenna array. The antenna array can be a virtual array created through the use multi-input multi-output (MIMO) technology.

FIGS. 1B and 1C illustrate an exemplary calibration system for a radar system. As discussed herein, the calibration system and radar system is first installed in a temporary installation. While in the temporary installation, the calibration system records calibration measurements. The calibration system is capable of recording a series of calibration measurements. Henceforth, an exemplary “measured channel response” refers to the data from these calibration measurements. As illustrated in FIGS. 1B and 1C, a radar 101 is mounted on top of an adjustable gimbal mount platform (hereinafter a “platform”) 120. The platform 120 is configured to rotate in one or both of azimuth (x-axis) and elevation (y-axis). The radar 101 is configured to transmit a signal to a reflecting object 103. FIGS. 1B and 1C illustrate a signal traversal path 104 extending from an array center 112 of an antenna array 110 to the reflecting object 103, while an expected path 105 is illustrated from the platform’s rotation center 122 to the reflecting object 103. The deviance in angle between the signal traversal path 104 and the expected path 105 causes a phase shift between the expected signal 105 and the actual reflected signal 104, as well as an error in the angle of arrival. This deviance is referred to as nodal displacement, and the phase shift is modeled as a direction dependent phase variation. Nodal displacement occurs for multiple of reasons. First, the height of the radar 101 on the platform 120 may not exactly match the plane of

the nodal point of rotation. Second, the nodal point may not exactly match the virtual center of the antenna array. The radar 101 may also have multiple antenna configurations with different virtual centers, and physical relocation of the radar system 101 may not be feasible. Last, there can be an error in estimating the correct nodal point.

FIG. 2 illustrates an exemplary radar/calibration system 201 which records calibration measurements while the radar/calibration system 201 is installed in a final platform or rotatable gimbal (the “platform”) 220. As illustrated in FIG. 2, the radar/calibration system 201 is mounted in the platform 220. A reflecting object 203 is positioned in front of the radar/calibration system 201. FIG. 2 illustrates a signal traversal path 204 extending from an array center 212 of an antenna array 210 of the radar 201 to a reflecting object 203. An expected path 205 is also illustrated extending from the rotation axis 222 of the platform 220 to the reflecting object 203. Rotation is achieved by the mechanics of the platform 220 itself. As in the previous paragraph, a direction dependent phase variation occurs due to nodal displacement when the rotation axis 222 of the platform 220 does not match the array center 212 of the antenna array 210.

FIG. 3A illustrates an exemplary radar using the calibration method and calibration system described in the current invention with at least one antenna 302 that is time-shared between at least one transmitter 306 and at least one receiver 308 via at least one duplexer 304. Output from the receiver(s) 308 is received by a control and processing module 310 that processes the output from the receiver(s) 308 to produce display data for the display 312. The control and processing module 310 is also operable to produce a radar data output that is provided to other control and processing units. The control and processing module 310 is also operable to control the transmitter(s) 306 and the receiver(s) 308.

FIG. 3B illustrates an alternative exemplary radar using the calibration method and system described in the current invention with separate sets of transmitter and receiver antennas. As illustrated in FIG. 3B, at least one antenna 302A for the at least one transmitter 306 and at least one antenna 302B for the at least one receiver 308.

FIG. 4 illustrates an exemplary MIMO (Multi-Input Multi-Output) radar 400 that is configured to use the calibration method and system described herein. With MIMO radar systems 400, each transmitter signal is rendered distinguishable from every other transmitter signal by using appropriate differences in the modulation, for example, different digital code sequences. Each receiver 408 correlates with each transmitter signal, producing a number of correlated outputs equal to the product of the number of receivers 408 with the number of transmitters 406 (virtual receivers= $RX_N * TX_N$). The outputs are deemed to have been produced by a number of virtual receivers, which can exceed the number of physical receivers 408.

FIG. 4 illustrates a radar system 400 with a plurality of antennas 402 connected to a plurality of receivers 408, and a plurality of antennas 404 connected to a plurality of transmitters 406. The radar system 400 of FIG. 4 is also a radar-on-chip system 400 where the plurality of receivers 408 and the plurality of transmitters 406, along with any processing to produce radar data output and any interface (like Ethernet, CAN-FD, Flex Ray etc.), are integrated on a single semiconductor IC (Integrated Circuit). Using multiple antennas allows the radar system 400 to determine the angle of objects/targets in the environment. Depending on the geometry of the antenna system 402, 404, different angles (e.g., with respect to the horizontal or vertical) can be determined. The radar system 400 may be connected to a

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network via an Ethernet connection or other types of network connections 414. The radar system 400 may also include memory 410, 412 to store software used for processing the received radio signals to determine range, velocity, and location of objects/targets in the environment. Memory may also be used to store information about objects/targets in the environment.

In practice, antenna elements have a directional gain and phase response. This response varies with respect to azimuth and elevation. The combination of transmitter and receiver antenna responses can be modeled as a new virtual antenna response. This response causes a gain and phase variation from the ideal signals at the virtual receivers. This effect can be divided into a per channel gain, per channel phase, direction dependent gain, and direction dependent phase. 15

In practice, leakage exists between antenna elements due to coupling effects. This coupling occurs between both the signals at the TX antenna elements and the RX antenna elements. This causes a deviation in both the signals that are transmitted by the transmitters 406 of the radar system 400 and the signals that are received by the receivers 408 of the radar system 400. The combined effect of coupling at both the transmitter and the receiver is modeled as coupling between virtual receivers. FIG. 5 illustrates the coupling in a virtual array. FIG. 5 illustrates a virtual antenna element array 501, a propagation front 502 of a far-field signal, and the path 503 of the signals to the virtual receivers. The signals at each virtual antenna element will couple. This coupling causes a gain and phase variation from the ideal signal at the virtual receivers. This impairment is henceforth referred to as mutual coupling. 20 25 30

In the preferred embodiment, the measured channel response is collected using a PMCW radar. Alternative embodiments may include other radar types.

Using the radar calibration systems described either in FIG. 1 or 2, one method of collecting the calibration measurements is a stop and go sweep. In this method, the radar system is rotated to the exact desired azimuth and elevation angles, where it stops before collecting the radar data. This method provides increased accuracy. 35 40

A second method of collecting the calibration measurements is a continuous sweep. In this second method, the radar system rotates in a continuous fashion and collects radar data while rotating. This method provides increased speed. However, it sacrifices accuracy due to angular smearing of the target response. There is no doppler impact since the rotation causes the effective target movement to be tangential to the radar. FIG. 6 illustrates exemplary sweep patterns. FIG. 6 illustrates azimuth sweeps 601 and elevation sweeps 602. The quantity, speed, and angular range of the sweeps is variable and chosen dependent on the array design. All sweeps contain a stationary measurement at boresight of the radar. 45 50

FIG. 7 illustrates the steps to an exemplary radar system calibration procedure. In step 701, an exemplary measurement collection process is carried out. In step 702, an exemplary phase correction process is carried out, which estimates and corrects for the per channel phase variation and direction dependent phase variation in the collected data. In step 703, an exemplary gain correction process is carried out, which estimates and corrects for the per channel gain variation and direction dependent gain variation in the phase-corrected data. In step 704, an exemplary ideal response refinement process is carried out, which uses the gain- and phase-corrected data to improve the angle-of-arrival estimation for the collected radar data. In step 705, an exemplary cross-coupling calibration process is carried out, 55 60 65

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which estimates and corrects for the cross-coupling effects remaining in the gain- and phase-corrected data.

The radar data is described by the following exemplary mathematical model. Denoting az and el as the azimuth and elevation angles (in radians) to the target, define the u-v space as:

$$u = \sin(az) \cos(el)$$

$$v = \sin(el)$$

Assuming a planar antenna array where the k^{th} (out of N_{vrx}) virtual antenna is located at $(0, dy_k, dz_k)$ in rectangular coordinates, the ideal receive data in the absence of any cross-coupling and no gain/phase variation is given by:

$$y_{ideal}(k, u, v) = e^{-j\frac{2\pi}{\lambda}(dy_k u + dz_k v)}$$

This ideal response of the N_{vrx} virtual antennas corresponding to a far-field target in the u and v (or equivalently in az and el) space is expressed in vector form as:

$$\vec{y}_{ideal}(u, v) = [y_{ideal}(0, u, v), y_{ideal}(1, u, v), \dots, y_{ideal}(N_{vrx} - 1, u, v)]^T$$

In the presence of cross-coupling, the received signal vector is $\vec{x} = A \vec{y}_{ideal}$, where $A = \{a_{m,k}\}$, $0 \leq m \leq N_{vrx} - 1$ is a matrix that captures both coupling and per channel gain and phase variation. With this impairment, the received data becomes:

$$x(k, u, v) = \sum_{m=0}^{N_{vrx}-1} \alpha_{m,k} e^{-j\frac{2\pi}{\lambda}(dy_m u + dz_m v)}$$

The vector representation of the channel response $\vec{x}(u, v)$ is then:

$$\vec{x}(u, v) = [x(0, u, v), x(1, u, v), \dots, x(N_{vrx} - 1, u, v)]^T$$

The data model described above applies to a far-field target. The embodiments of the method and calibration system discussed herein equally applies to a near-field target as well with a corresponding modification of the signal vectors defined above. The data model can be updated for non-nodal displacement for the radar in the data collection setup as follows:

$$x_{meas}(k, u, v) = \gamma(u, v) \sum_{m=0}^{N_{vrx}-1} \alpha_{m,k} e^{-j\frac{2\pi}{\lambda}(dy_m(u - \delta u(u, v)) + dz_m(v - \delta v(u, v)))}$$

Here, $\gamma(u, v)$ is due to the angle dependent phase correction (e.g., as a result of nodal displacement). $\delta u(u, v)$ and $\delta v(u, v)$ represent the angle dependent (hence the notation that these parameters are dependent on the

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angle of incidence as well) mismatch between the expected direction and the actual sampled direction.

The vector representation $\vec{x}_{meas}(u, v)$ is:

$$\vec{x}_{meas}(u, v) = [x_{meas}(0, u, v), x_{meas}(1, u, v), \dots, x_{meas}(N_{rx} - 1, u, v)]^T$$

or

$$\vec{x}_{meas}(u, v) = \gamma(u, v) A y_{ideal}(u - \delta u(u, v), v - \delta v(u, v))$$

FIG. 8 illustrates an exemplary calibration procedure for direction dependent phase variation and per channel phase variation. In step 801, the estimates of the direction dependent phase variation and per channel phase variation are initialized to zero. Then, in step 802, a correction term is computed as the normalized complex conjugate of the measured channel response at boresight of the radar system. Then an iterative procedure begins. In step 803, the direction dependent phase variation is estimated using least-squares to minimize the difference between the corrected channel response and the ideal channel response. In step 804, the channel response is corrected again with this phase. Next in step 805, the per channel phase variation is estimated using least-squares to minimize the difference between the corrected channel response and the ideal channel response, now across all directions. In step 806, the channel response is corrected again with this phase. This iterative procedure is repeated for a fixed number of iterations or until convergence.

This phase calibration procedure can be described mathematically using the previous exemplary signal model. The initial coupling matrix in step 802 is set to zeros, except for the diagonal elements which are initially set to

$$\alpha_{k,k}^{brs} = \frac{x_{meas}^*(k, 0, 0)}{|x_{meas}(k, 0, 0)|}$$

since $\vec{x}_{meas}(k, 0, 0)$ is the channel measured at boresight on the k^{th} virtual element. Accordingly with step 801 and step 802, we now initialize the following terms: direction dependent phase term: $\angle \tilde{\gamma}^0(u, v) = 0$, per channel phase term: $\angle \tilde{\alpha}_{k,k}^0 = 0$, and the array response corrected for the direction dependent and per channel phase terms $\tilde{x}^0(k, u, v) = \alpha_{k,k}^{brs} x_{meas}(k, u, v)$. Then the iterative procedure begins. The superscript it is the iteration index. The direction-dependent least squares solution, $\angle \tilde{\gamma}^{it}(u, v)$, in step 803 is obtained by minimizing the cost function below:

$$C_{1,phase}(u, v) = \sum_{k=0}^{N_{rx}-1} |y_{ideal}(k, u, v) e^{-j\angle \tilde{\gamma}^{it}(u, v)} - \tilde{x}^{it-1}(k, u, v)|^2$$

The radar data is then updated in step 804 as

$$\tilde{x}^{it}(k, u, v) = \tilde{x}^{it-1}(k, u, v) e^{j\angle \tilde{\gamma}^{it}(u, v)}$$

The per channel least squared solution, $\angle \tilde{\alpha}_{k,k}^{it}$, in step 805 is obtained by minimizing the cost function below

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$$C_{2,phase}(k) = \sum_{u,v} |y_{ideal}(k, u, v) e^{-j\angle \tilde{\alpha}_{k,k}^{it}} - \tilde{x}^{it}(k, u, v)|^2$$

The radar data is then updated in step 806 as

$$\tilde{x}^{it}(k, u, v) = \tilde{x}^{it}(k, u, v) e^{j\angle \tilde{\alpha}_{k,k}^{it}}$$

This procedure loops for a finite number of iterations. Let the number of iterations be L. At the end of iterations, we obtain the following information: updated virtual array response (corrected for phase which corrects for nodal displacement as well as phase response per angle) $\tilde{x}(k, u, v) = \tilde{x}^L(k, u, v)$, estimate of direction dependent phase correction

$$\angle \tilde{\gamma}(u, v) = \sum_{it=1}^L \angle \tilde{\gamma}^{it}(u, v),$$

and estimate of per channel phase variation

$$\angle \tilde{\alpha}_{k,k} = \sum_{it=1}^L \angle \tilde{\alpha}_{k,k}^{it}.$$

FIG. 9 illustrates an exemplary calibration procedure for direction dependent gain variation and per channel gain variation. First in step 901, the estimates of the direction dependent gain and per channel gain are initialized to unity. The corrected channel response after the phase correction is now used. Then an iterative procedure begins. In step 902, the direction dependent gain is estimated using least-squares to minimize the difference between the corrected channel response and the ideal channel response. Note that the ideal channel response is unity across all virtual receivers. In step 903, the channel response is corrected with this gain. Next in step 904, the per channel gain is estimated using least-squares to minimize the difference between the corrected channel response and the ideal channel response, now across all directions. In step 905, the channel response is corrected again with this gain. This iterative procedure is repeated for a fixed number of iterations or until convergence.

This gain calibration procedure can be described mathematically using the previous exemplary signal model. Accordingly, with step 901, the following terms are initialized: direction dependent amplitude term: $|\tilde{\gamma}^0(u, v)| = 1$, per channel amplitude term: $|\tilde{\alpha}_{k,k}^0| = 1$, and the array response as corrected at the output of the previous phase calibration stage $\tilde{x}^0(k, u, v) = \tilde{x}(k, u, v)$. Then the iterative procedure begins. The direction-dependent least-squares solution, $|\tilde{\gamma}^{it}(u, v)|$, in step 902, is obtained by minimizing the cost function below:

$$C_{1,gain}(u, v) = \sum_{k=0}^{N_{rx}-1} ||y_{ideal}(k, u, v)| |\tilde{\gamma}^{it}(u, v)| - |\tilde{x}^{it-1}(k, u, v)||^2$$

The radar data is then updated in step 903 as:

$$\tilde{x}^{it}(k, u, v) = \tilde{x}^{it-1}(k, u, v) |\tilde{\gamma}^{it}(u, v)|$$

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The per channel least-squares solution, $|\tilde{a}_{k,k}^{it}|$, in step **904** is obtained by minimizing the cost function below

$$C_{2,gain}(k) = \sum_{u,v} |y_{ideal}(k, u, v) |\tilde{a}_{k,k}^{it}| - \tilde{x}^{it}(k, u, v)|^2$$

The radar data is then updated in step **905** as:

$$\tilde{x}^{it}(k, u, v) = \tilde{x}^{it}(k, u, v) |\tilde{a}_{k,k}^{it}|$$

This procedure loops for a finite number of iterations. Let the number of iterations be L. At the end of iterations, we obtain the following information: updated virtual array response (corrected for direction dependent phase which corrects for nodal displacement as well as amplitude/phase response per angle) $\tilde{x}(k, u, v) = \tilde{x}^L(k, u, v)$, an estimate of direction dependent amplitude correction,

$$|\tilde{y}(u, v)| = \prod_{i=1}^L |\tilde{y}^{it}(u, v)|,$$

and an estimate of per channel amplitude variation

$$|\alpha_{k,k}| = \prod_{i=1}^L |\alpha_{k,k}^{it}|.$$

The total per channel gain and phase correction terms are combined into a matrix, which is referred to as the diagonal antenna correction matrix. Using the exemplary mathematical model, this diagonal antenna correction matrix is defined as:

$$\tilde{A}_d^{-1} = \text{diag}\{|\tilde{a}_{k,k}| e^{jL\tilde{\alpha}_{k,k}}\},$$

$$0 \leq k \leq N_{vrx} - 1$$

In an embodiment of the method and calibration system discussed herein, the ideal channel response can be refined to correct for setup error and nodal displacement error after the diagonal antenna correction. The MUSIC algorithm is used, which exploits knowledge of the number of objects to provide a super-resolution estimate of the actual object directions in each measurement. These directions are then used to recompute the ideal channel response. This refined ideal channel response is used during the cross-coupling calibration process.

FIG. 10 illustrates the calibration procedure for cross-coupling. First in step **1001**, a cross coupling estimate matrix is initialized to zeros. Then in step **1002**, a virtual channel is selected. In step **1003**, a list of retired cross channels is initialized to empty. Then in step **1004**, the measured response of the selected virtual channel is projected onto the ideal response for all cross channels. In step **1005**, the cross channel that corresponds to the maximum of the projection and is not yet retired is selected. The value of this projection is also recorded. In step **1006**, the contribution of the selected cross channel is removed from the measured response of the selected virtual channel. In step **1007**, if the contribution is greater than a given threshold, the cross-coupling matrix element corresponding to the selected vir-

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tual channel and the selected cross channel is set to the recorded projection value in step **1008**. Then in step **1009**, the selected cross channel is retired. The process repeats at the projection in step **1004**. If the contribution was not greater than a given threshold in step **1007**, then the process repeats at the virtual channel selection in step **1002** with the next virtual channel. This iterates until the process has completed for all virtual channels, as shown by step **1010**. In step **1011**, the cross-coupling matrix is then normalized by multiplying by the square root of the number of virtual antennas divided by the L2 norm squared of the diagonal elements of the cross-coupling matrix. The cross-coupling correction matrix is then estimated using the inverse of this cross-coupling matrix.

This cross-coupling calibration procedure can be described mathematically using the previous exemplary signal model. First in step **1001**, the cross-coupling matrix $\tilde{A}_c = 0_{N_{vrx} \times N_{vrx}}$ is initialized. Then the iterative procedure begins for each virtual channel. In step **1002**, the index of the current virtual channel is denoted as k. In step **1003**, initialize the list of retired cross channels $S = \{\}$ is initialized. The projection of the k virtual channel response onto the ideal response for all cross channels in step **1004** is given by:

$$\beta_{k,m} = \sum_{u,v} \tilde{x}(k, u, v) y_{ideal}(m, u, v),$$

$$0 \leq m \leq N_{vrx} - 1$$

Then in step **1005**, the channel, m_{max} , with the largest of $\beta_{k,m}$ is found as:

$$m_{max} = \underset{m \notin S}{\text{argmax}} |\beta_{k,m}|$$

If this is the first iteration, $|\beta_{k,m_{max}}|$ is recorded as the largest coupling value, β_{max} . In this implementation of the algorithm, these values are used in the thresholding function in step **1007**. Then in step **1006**, the contribution to the measured channel response is removed from the selected cross-channel from above as:

$$\tilde{x}(k, u, v) = \tilde{x}(k, u, v) - y_{ideal}(m_{max}, u, v) \beta_{k,m_{max}}$$

In step **1007**, if the ratio between $|\beta_{k,m_{max}}|$ and β_{max} exceeds a certain threshold, Th_{cpl} , m_{max} is added to S in step **1009**, and $\tilde{A}_c(k, m_{max})$ is set to $\beta_{k,m_{max}}$ in step **1008**, and then the process goes to the projection in step **1004**. If the threshold test in step **1007** fails, the search for the k^{th} virtual channel is ended when the next virtual channel is repeated at step **1002**. Once the iterations are completed over all the virtual channels as indicated by step **1010**, the estimated cross-coupling matrix is normalized in step **1011** as follows:

$$\tilde{A}_c = \frac{\sqrt{N}}{\|\text{diag}(\tilde{A}_c)\|_2} \tilde{A}_c$$

A final correction matrix is computed through matrix multiplication of the cross-coupling correction matrix and the diagonal antenna correction matrix. Using the previous

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exemplary signal model, this correction matrix is defined mathematically as $\tilde{C} = A_c^{-1} \tilde{A}_d^{-1}$. This final correction matrix is implemented into the radar processing. The vector of data received at the virtual receivers is multiplied by this correction matrix before being multiplied by a steering matrix to achieve the calibrated beamformed output. The steering matrix is a stack of steering vectors, whose elements correspond to the desired complex beamforming weights. In the preferred embodiment, these vectors are the complex conjugate of the ideal channel response for the antenna array at a desired set of directions.

Thus, the exemplary embodiments discussed herein provide for the calibration of a radar system that corrects or adjusts for a misalignment between an array center of an antenna array of the radar system and a rotation center of the radar system via a platform of a calibration module that rotates the radar system in both azimuth and elevation. The calibration module calculates a misalignment between the array center of the antenna array and the rotation center of the radar system. At least one receiver of the radar system corrects for phase distortion and angle-of-arrival error due to the calculated misalignment. The misalignment between the array center of the antenna and the rotation center of the platform is a nodal displacement.

Changes and modifications in the specifically described embodiments can be carried out without departing from the principles of the present invention, which is intended to be limited only by the scope of the appended claims, as interpreted according to the principles of patent law including the doctrine of equivalents.

The invention claimed is:

1. A radar system comprising:

- a transmitter communicatively coupled to a transmitter antenna array, wherein the transmitter is configured to transmit radio signals via the transmitter antenna array;
- a receiver communicatively coupled to a receiver antenna array, wherein the receiver is configured to receive radio signals via the receiver antenna array that include radio signals transmitted by the transmitter and reflected from objects in an environment;
- a calibration module comprising a platform configured to rotate the transmitter antenna array and the receiver antenna array in azimuth and/or elevation, wherein respective array centers of the transmitter antenna array and/or the receiver antenna array are not aligned with the platform's rotation center;

wherein the calibration module is configured to access receiver data from the receiver for each of a plurality of azimuth and/or elevation angles as the platform is rotated, wherein the calibration module is operable to calculate a misalignment value from the receiver data to define a misalignment between the array center of the receiver antenna array and the rotation center of the platform, and wherein the receiver is operable to perform antenna calibrations that includes accounting for the misalignment based upon the calculated misalignment value.

2. The radar system of claim 1, wherein the platform is configured to rotate in discrete steps, and wherein the calibration module is operable to collect receiver data from the receiver at each discrete step.

3. The radar system of claim 2, wherein the calibration module is configured to rotate the platform to discrete angles before collecting receiver data.

4. The radar system of claim 2, wherein the platform is configured to rotate discretely for azimuth sweeps and/or elevation sweeps.

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5. The radar system of claim 1, wherein the platform is configured to rotate in a continuous sweep of angles, and wherein the calibration module is operable to collect receiver data while the platform is rotating.

6. The radar system of claim 5, wherein the platform is configured to rotate continuously for azimuth sweeps and/or elevation sweeps.

7. The radar system of claim 1, wherein the receiver is operable to perform the antenna calibrations to account for phase distortion and angle-of-arrival error.

8. The radar system of claim 1 further comprising a plurality of transmitters and a plurality of receivers, wherein each transmitter of the plurality of transmitters is communicatively coupled to the transmitter antenna array, and wherein each receiver of the plurality of receivers is communicatively coupled to the receiver antenna array.

9. The radar system of claim 8, wherein the calibration module is configured to access receiver data from each receiver of the plurality of receivers for each of a plurality of azimuth and/or elevation angles as the platform is rotated, and wherein the calibration module is configured to calculate a respective misalignment value for each receiver of the plurality of receivers.

10. The radar system of claim 9, wherein each receiver of the plurality of receivers is operable to perform antenna calibrations to account for the respective phase distortion and angle-of-arrival errors that includes accounting for the misalignment based upon the calculated misalignment values of each respective receiver of the plurality of receivers.

11. The radar system of claim 10, wherein the misalignment between the array center of the receiver antenna array and the rotation center of the platform is a nodal displacement, and wherein the array center of the receiver antenna array is a nodal point.

12. The radar system of claim 11, wherein the calibration module is operable to process phase distortion and angle-of-arrival error measurements into a correction matrix to calibrate for transmitter and/or receiver impairments which include at least one of phase error due to nodal displacement, per channel phase variation, direction dependent phase variation, per channel amplitude variation, direction dependent amplitude variation, and channel response cross-coupling.

13. The radar system of claim 12, wherein the calibration module is operable to modify its measurement, collection, and calibration processing to optimize different objective functions including at least one of speed and manner of rotation, quantity of measurements collected, and the selection of antenna(s) and channel(s) transmitting and receiving the signals, and wherein these modifications also include parameters in the processing that controls the computation of the correction matrix and affect the processing speed and correction accuracy.

14. The radar system of claim 8 further comprising an antenna switch, wherein the transmitter antenna array and the receiver antenna array each comprise multiple antennas, and wherein each transmitter of the plurality of transmitters and each receiver of the plurality of receivers are coupled to the corresponding multiple transmitter antennas and multiple receiver antennas, respectively, via the antenna switch.

15. A radar system comprising:

- a plurality of transmitters, each communicatively coupled to a transmitter antenna array, wherein each transmitter of the plurality of transmitters is configured to transmit radio signals via the transmitter antenna array;
- a plurality of receivers, each communicatively coupled to a receiver antenna array, wherein each receiver of the

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plurality of receivers is configured to receive radio signals via the receiver antenna array that include radio signals transmitted by the transmitters and reflected from objects in an environment;

a calibration module comprising a platform configured to rotate the transmitter antenna array and the receiver antenna array in azimuth and/or elevation, wherein respective array centers of the transmitter antenna array and the receiver antenna array are not aligned with the platform's rotation center;

wherein the calibration module is configured to access receiver data from each receiver of the plurality of receivers for each of a plurality of azimuth and/or elevation angles as the platform is rotated, wherein the calibration module is operable to calculate a respective misalignment value for each receiver of the plurality of receivers from the receiver data to define a misalignment between the array center of the receiver antenna array and the rotation center of the platform.

16. The radar system of claim 15, wherein each receiver of the plurality of receivers is operable to perform antenna calibrations to account for respective phase distortion and angle-of-arrival errors that includes accounting for the misalignment based upon the calculated misalignment values of each respective receiver of the plurality of receivers.

17. The radar system of claim 16, wherein the misalignment between the array center of the receiver antenna array and the rotation center of the platform is a nodal displacement, and wherein the array center of the receiver antenna array is a nodal point.

18. The radar system of claim 17, wherein the calibration module is operable to process phase distortion and angle-of-arrival error measurements into a correction matrix to calibrate for transmitter and/or receiver impairments which include at least one of phase error due to nodal displacement, per channel phase variation, direction dependent phase variation, per channel amplitude variation, direction dependent amplitude variation, and channel response cross-coupling.

19. The radar system of claim 18, wherein the calibration module is operable to modify its measurement, collection, and calibration processing to optimize different objective functions including at least one of speed and manner of rotation, quantity of measurements collected, and the selection of antenna(s) and channel(s) transmitting and receiving the signals, and wherein these modifications also include parameters in the processing that controls the computation of the correction matrix and affect the processing speed and correction accuracy.

20. A method for calibrating a radar system for system impairments, wherein the method comprises:

transmitting, with a transmitter, radio signals;

receiving, with a receiver, radio signals that include radio signals transmitted by the transmitter and reflected from objects in an environment;

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wherein the transmitter and the receiver are coupled to an antenna array;

rotating, with a platform, the antenna array in both azimuth and elevation, and wherein an array center of the antenna array is not aligned with the platform's rotational center;

in the presence of at least one object, collecting from the receiver data defined by received radio signals reflected from the at least one object at selected azimuth and/or elevation angles, and calculating a misalignment value for a misalignment between the array center of the antenna array and the rotation center of the platform, and

correcting, with the receiver, antenna calibration errors that accounts for the misalignment based upon the misalignment value, wherein the misalignment between the array center of the antenna and the rotation center of the platform is a nodal displacement, and wherein the array center of the antenna array is a nodal point.

21. The method of claim 20, wherein correcting for antenna calibration errors accounts for phase distortions and angle-of-arrival errors.

22. The method of claim 21 further comprising processing phase distortion and angle-of-arrival error measurements into a correction matrix to calibrate for transmitter and/or receiver impairments, which include at least one of phase error due to nodal displacement, per channel phase variation, direction dependent phase variation, per channel amplitude variation, direction dependent amplitude variation, and channel response cross-coupling.

23. The method of claim 22 further comprising solving phase error and amplitude variations via an iterative least squares optimization solution.

24. The method of claim 21 further comprising estimating, with the receiver, angles-of-arrival of the collected reflected signals or determining angles-of-arrival of the collected reflected signals based upon prior knowledge of the at least one object's location relative to the receiver.

25. The method of claim 21 further comprising modifying measurement, collection, and calibration processing to optimize different objective functions including at least one of speed and manner of rotation, quantity of measurements collected, and the selection of antenna(s) and channel(s) transmitting and receiving the signals, and wherein these modifications also include parameters in the processing that controls the computation of the correction matrix and affect the processing speed and correction accuracy.

26. The method of claim 20, wherein the platform is rotated in either a continuous manner or in discrete steps, wherein, when rotating in discrete steps, the platform is rotated to discrete angles before collecting receiver data, and wherein, when rotating continuously, the platform is rotated in a continuous sweep of angles while collecting receiver data.

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